Downside Deviation

In practice, we may be concerned with values of return (or another variable) below some level other than the mean. For example, if our return objective is 6.0 percent annually (our minimum acceptable return), then we may be concerned particularly with returns below 6.0 percent a year. The target downside deviation, also referred to as the **target semideviation**, is a measure of dispersion of the observations (here, returns) below a target—for example 6.0 percent. To calculate a sample target semideviation, we first specify the target. After identifying observations below the target, we find the sum of those squared negative deviations from the target, divide that sum by the total number of observations in the sample minus 1, and, finally, take the square root.

Sample Target Semideviation Formula. The target semideviation, s_{Target}, is:

$$s_{Target} = \sqrt{\sum_{\text{for all} X_i \le B}^{n} \frac{(X_i - B)^2}{n - 1}}, \tag{6}$$

where B is the target and n is the total number of sample observations. We illustrate this in Example 3.

EXAMPLE 3

Calculating Target Downside Deviation

Consider the monthly returns on a portfolio as shown in Exhibit 11:

Exhibit 11: Monthly Portfolio Returns

Month	Return (%)
January	5
February	3
March	−1
April	−4
May	4
June	2
July	0
August	4
September	3
October	0
November	6
December	5

1. Calculate the target downside deviation when the target return is 3 percent.

   Solution:

Month	Observation	Deviation from the 3% Target	Deviations below the Target	Squared Deviations below the Target
January	5	2	—	—
February	3	0	—	—